

# Candidate Assessment

## Role: Data Engineer – IFTA Evidence Data Engineering

This assessment is designed with an AWS-based IFTA audit environment in mind. Candidates are therefore encouraged to demonstrate their solution using relevant AWS services (e.g., Amazon S3, RDS, Redshift, Textract, SageMaker, Lambda, Glue, Athena).

### About IFTA

The International Fuel Tax Agreement (IFTA) is an agreement amongst Canadian and United States jurisdictions that simplifies fuel tax reporting for motor carriers operating across multiple jurisdictions. IFTA compliance is based on reconciling fuel purchase records with distance traveled by jurisdiction. While IFTA defines the data elements required for audits, it does not prescribe a standard format for how supporting evidence is submitted. As a result, audit evidence is often provided in varied and inconsistent formats, such as paper receipts, scanned documents, PDFs, spreadsheets, and telematics exports. Accurate extraction, normalization, and end-to-end traceability of this evidence are therefore critical to enable reliable analytics and effective audit review.

### Context

You are provided with anonymized IFTA evidence files, including:

- Fuel invoices (scanned images / documents with varying formats)
- Distance logs (structured log of trips with odometer readings and distances – excel and scanned images)

These files reflect real-world IFTA evidence characteristics:

- inconsistent layouts
- missing or ambiguous fields
- data quality issues that should be handled explicitly

### Objective

Design and implement an AWS-based data engineering solution that:

- securely ingests the evidence files
- extracts structured data from invoices and logs
- normalizes the data into analytics-ready tables
- tracks lineage and extraction confidence
- produces storage layers suitable for analytics and machine learning workloads

### Scope of Work

#### 1. Secure Data Ingestion (AWS)

- Ingest the invoice documents and distance logs into appropriate storage layer
- Implement storage architecture with clear zones

- Apply encryption (SSE-KMS) and least-privilege IAM access

## 2. Data Extraction

From the provided evidence files, extract fields such as:

- invoice / receipt number
- transaction date
- transaction time
- vendor name, location (city, province)
- fuel type
- litres purchased
- price per litre and total cost
- tax breakdown (GST / HST / PST)
- form of payment
- tax indicators
- trip date
- trip origin and destination
- trip start and end time
- start and end odometer
- distance (km)
- VIN or truck number
- Distance by jurisdiction

For extracted data, capture:

- source file reference
- page or image reference (for invoices)
- extraction confidence score

## 3. Normalization Layer

Design a normalized data model, for example:

- fuel purchase facts
- trip distance facts
- date, location, and jurisdiction dimensions

Explain how your design supports:

- aggregation by jurisdiction and period
- reconciliation between fuel and distance evidence

## 4. Lineage & Data Quality

- Ensure every curated record can be traced back to:
  - original file
  - extraction output
- Explicitly flag (do not silently fix):
  - missing values

- inconsistent distances
- implausible fuel patterns

## 5. Storage Ready for Advanced Analytics

- Store curated datasets in appropriate format (e.g., Parquet)
- Partition appropriately for query efficiency
- Register schemas for analytics access (e.g., Glue / Athena)

## Deliverables

- Code, notebooks, or clear pseudocode
- AWS storage architecture overview
- Normalized data model description
- Sample curated outputs
- Summary of data quality issues and assumptions

## Interview Presentation (maximum 20 minutes)

You will walk the interviewers through:

- problem understanding
- approach and workflow
- AWS architecture
- extracted data, quality issues and suggestions to improve
- a live demonstration of what you built and which AWS services you used.

## Evaluation Rubric (100 points total)

| Category                              | Points | What We Look For                              |
|---------------------------------------|--------|-----------------------------------------------|
| Problem understanding                 | 15     | IFTA compliance, purpose & audit needs        |
| Data extraction, processing & quality | 20     | Use of technology, confidence, accuracy, gaps |
| AWS architecture                      | 20     | Clarity, automation potential, auditability   |
| Code & reproducibility                | 15     | Organization, clarity and ability to re-run   |
| Communication & presentation          | 30     | Narrative and decision making                 |